# DECOHERENCE

**Project Report**

*Submitted in partial fulfilment of the requirements for the award of Degree of*

**Bachelor of Technology**
*in*
**Engineering Physics**

*by*

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# CERTIFICATE

*This is to certify that the report titled* ***"Decoherence"*** *is a bonafide record of the project done by* ***Yogeshwaran R*** *(**B200161EP**) under my supervision and guidance, in partial fulfilment of the requirements for the award of Degree of* ***Bachelor of Technology*** *in* ***Engineering Physics*** *from* ***National Institute of Technology Calicut*** *for the year 2023-24.*

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# Declaration

I declare that this project report titled **"Decoherence"** submitted in partial fulfilment of the degree of B.Tech in Engineering Physics is a record of original work carried out by me under the supervision of **Dr. Sreeraj T.P** and has not formed the basis of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice of reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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# 1 Introduction

What we experience in this macroscopic world is a result of microscopic objects that are governed by quantum theory. This theory has some exceptional features like quantum entanglement and quantum superpositions which led to lots of confusion for decades. They are the reasons why quantum theory seems to be in conflict with our intuitions of our macroscopic (classical) world. The classical macroscopic world does not show any of these quantum features. However, a crucial question emerges: How can quantum physics give rise to the classical world? (Quantum to classical transitions). Decoherence is one of the possible answers that we will attempt to explore here.

Basically, from the ideas of classical physics of idealised isolated systems we had incorporated into quantum mechanics easily without much scrutiny. However, isolated quantum systems hinder us from understanding the quantum-to-classical transition. We therefore require open quantum systems which interact with the environment not just as a fix but an essential and more complete description of many systems of interest today, from quantum optics to quantum computing.

**Back to basics**

## 1.1 Density matrix and reduced density matrix

For the study of decoherence, density matrix formulation, especially, reduced density matrix plays an important role since system cannot be described by pure states due to the entanglement of system and environment.

Defining the density matrix for pure states $|\psi\rangle$ to be

$$\hat{\rho} = |\psi\rangle \langle\psi| \,. \tag{1}$$

Any state $|\psi\rangle$ can be expressed as a superposition of some orthonormal basis states $|\psi_i\rangle$

$$|\psi\rangle = \sum_i c_i |\psi_i\rangle \tag{2}$$

then the density matrix is

$$\hat{\rho} = \sum_{i,j} c_i c_j^* |\psi_i\rangle \langle\psi_j| \tag{3}$$

From here the off-diagonal ($i \neq j$) terms are the reasons for the interference effects. So, usually referred to as interference terms. Note the density operator $\rho$ can be diagonal or not based on the basis we choose to represent them. With this in hands the quantity of interest is the expectation values of hermitian operator ($\hat{O}$) which can be obtained by

$$Tr(\hat{\rho}\hat{O}) = \sum_i \langle o_i| (|\psi\rangle \langle\psi|)\hat{O} |o_i\rangle \tag{4}$$

here $|o_i\rangle$ are eigenstates of $\hat{O}$

$$= \sum_i o_i| \langle o_i|\psi\rangle |^2 \tag{5}$$

$$= \langle\hat{O}\rangle . \tag{6}$$

We have seen that $\hat{\rho}$ is completely equivalent to state $|\psi\rangle$ and of not much use. But, $\hat{\rho}$ becomes handy when we insufficient information about the state of the system (mixed states). What it means is the classical ensemble of set of possible pure states $|\psi_i\rangle$ with probabilities $p_i's$, then this density matrix is given by

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle\psi_i| \tag{7}$$

with $p_i's \geq 0$ and $\sum_i p_i = 1$. This now is a classical probability distribution of pure-state density matrices $\hat{\rho}_i = |\psi_i\rangle \langle\psi_i|$.

**Reduced density matrix:** Consider two systems $\mathcal{S}$ and $\mathcal{A}$ now the reduced density matrix of system $\mathcal{S}$ is

$$\hat{\rho}_{\mathcal{S}} = Tr_{\mathcal{A}}(\hat{\rho}). \tag{8}$$

where $\hat{\rho}$ is of composite system $\mathcal{S}$ and $\mathcal{A}$. And similarly expectation value of observables of system $\mathcal{S}$ can be derived from reduced density matrix as

$$\langle \hat{O}_{\mathcal{S}} \rangle = Tr_{\mathcal{S}}(\hat{\rho}_{\mathcal{S}} \hat{O}_{\mathcal{S}}). \tag{9}$$

# 2 The Measurement Problem in Quantum Mechanics

Nearly a century has passed since the foundation of quantum mechanics and despite this, even today, standard textbooks still do not address this measurement problem, and claiming that while physical quantities (observables) are represented by Hermitian operators, measuring them somehow instantly change the system's quantum state into one of those operator's eigenstates. But rather it involves stages of process.

And the measurement process itself consists of three different parts as mentioned below[1] :

1. *The problem of the preferred basis* : What chooses the preferred physical quantities (observables) to be observed in nature?
2. *The problem of the nonobservability of interference* : Why do quantum interference effects vanish at macroscopic level?
3. *The problem of outcomes* : What selects the particular outcome (eigenstate of the observable) of different outcomes available given by quantum probability distribution.

## 2.1 The von Neumann Measurement

John von Neumann's idea was to describe the measurement process in entirely quantum mechanical terms. This means that not only the measured system but also the apparatus should be treated as a quantum mechanical object. Note that this approach is in sharp contrast to the Copenhagen interpretation, where the

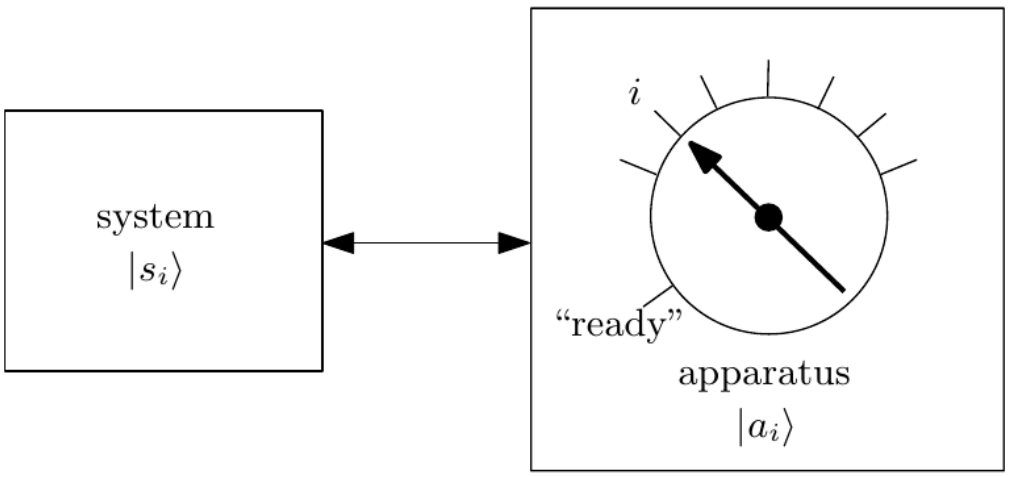


Figure 1: Illustration of establishment of one-to-one correspondence between the state of the system and the state of the apparatus [1].

measurement apparatuses are excluded from the quantum mechanical description and are postulated as intrinsically classical. This ideal quantum measurement is the easiest way to understand how quantum entanglement arises.

Let us consider a quantum mechanical system $\mathcal{S}$ in the Hilbert space $\mathcal{H}_{\mathcal{S}}$ with basis vectors $|s_i\rangle$, and a measuring apparatus $\mathcal{A}$ with basis vectors $|a_i\rangle$ in a Hilbert space $\mathcal{H}_{\mathcal{A}}$. The purpose of the apparatus is now to measure the state of the system $\mathcal{S}$. We can think of the apparatus as having some kind of pointer that moves to the position *'i'*, represented by the state $|a_i\rangle$, if the system is measured to be in the state $|s_i\rangle$ (the main idea here is to establish a one-to-one correspondence between the state of the system and the state of the apparatus). Initially the apparatus is in some initial "ready" state $|a_r\rangle$, the dynamical measurement interaction between the system and the apparatus will then be of the form

$$|s_i\rangle\,|a_r\rangle \;\longrightarrow\; |s_i\rangle\,|a_i\rangle \tag{10}$$

There is an assumption that the measurement interaction does not change the state of the system (ideal). Let us assume that the initial (more general) state of the system is in superposition of the

states $|s_i\rangle$

$$|\psi_s\rangle = \sum_i c_i |s_i\rangle \tag{11}$$

Then the initial state of the composite $\mathcal{S} + \mathcal{A}$ is given by

$$|\psi\rangle = \sum_i c_i |s_i\rangle |a_r\rangle \tag{12}$$

under the unitary evolution of the whole system the transformation must be of the form

$$\mathcal{U}(t) |s_i\rangle |a_r\rangle \longrightarrow |s_i\rangle |a_i\rangle \tag{13}$$

The total Hamiltonian of the system-apparatus combine as $H = H_{\mathcal{S}} + H_{\mathcal{A}} + H_{int}$ , and to satisfy (4) we can consider the interaction of the von Neumann form[2]

$$H_{int} = \sum_i |s_i\rangle \langle s_i| \otimes A_i \tag{14}$$

Here the $A_i's$ are operators acting on $\mathcal{H}_{\mathcal{A}}$, and during the measurement time we will consider that the interaction term in the Hamiltonian dominates over the other two terms ( so that $H \simeq H_{int}$). This leads to

$$e^{-iH_{int}t/\hbar}(\sum_i c_i |s_i\rangle |a_r\rangle) \tag{15}$$

$$= \sum_i c_i |s_i\rangle \otimes e^{-iA_i t/\hbar} |a_r\rangle \tag{16}$$

Let

$$|a_i\rangle = e^{-iA_i t/\hbar} |a_r\rangle$$

which implies that[1]

$$|\psi(t)\rangle = \sum_i c_i |s_i\rangle |a_i\rangle \tag{17}$$

Note that as the apparatus state entangles with the system the superposition is also carried on to the apparatus. And also the measurement is not complete yet, so often it is referred to as pre-measurement.

### 2.2 The problem of preferred basis

From the von Neumann scheme above the final state after the premeasurement can be expressed in many ways by the choice of the basis

$$\sum_i c_i \left|s_i\right\rangle \left|a_i\right\rangle = \sum_i c_i' \left|s_i'\right\rangle \left|a_i'\right\rangle \tag{18}$$

Now, for the detector to give us a classical measurement (macroscopically distinguishable) we require its basis states to be orthogonal $\langle a_i | a_j \rangle = 0$ for $i \neq j$.

Coming back to the question of the given infinite number of ways to express the states in different basis, how does it select an unique (preferred) basis? Let's see an example of the Stern-Gerlach experiment and infer from it. If we choose to measure spin in z-direction which fixes the basis of the apparatus to $\left|0_z\right\rangle_{\mathcal{A}}, \left|1_z\right\rangle_{\mathcal{A}}$ for example[3],

$$\left|\psi_f\right\rangle = \frac{1}{\sqrt{2}}(\left|z,+\right\rangle_{\mathcal{S}} \left|z,-\right\rangle_{\mathcal{A}} - \left|z,-\right\rangle_{\mathcal{S}} \left|z,+\right\rangle_{\mathcal{A}}) \tag{19}$$

or

$$\left|\psi_f\right\rangle = \frac{1}{\sqrt{2}}(\left|x,+\right\rangle_{\mathcal{S}} \left|x,-\right\rangle_{\mathcal{A}} - \left|x,-\right\rangle_{\mathcal{S}} \left|x,+\right\rangle_{\mathcal{A}}) \tag{20}$$

But this suggests that the apparatus $\mathcal{A}$ could simultaneously measure the two non-commuting operators $\hat{\sigma_x}$ and $\hat{\sigma_z}$ , which is impossible as the apparatus can only be set up one way at a time.

Thus, the selection of the configuration of $\mathcal{A}$ uniquely determines the basis of $\mathcal{S}$ that we observe. Furthermore, this can be extended beyond spin directions to any non-commuting values. This solves the problem of preferred basis.

## 3 Decoherence

### 3.1 Basic concept of decoherence

Decoherence is the most commonly used approach to tackle the problem of *quantum to classical transition.* As the theory of ide-

alised isolated system had great success in classical physics it got incorporated into quantum mechanics easily without much scrutiny. But the *key idea* of the decoherence theory is that no system can be perfectly isolated[4]. When the system interacts with the environment (have a very large number of degrees of freedom, inaccessible for all practical purposes), it monitors the macroscopic system and therefore locally suppresses interference of macroscopically distinct states. Thus we have to study it under open quantum systems.

### 3.2 Environment-Induced Decoherence

Let's consider a system of interest $\mathcal{S}$ interacting with its environment $\mathcal{E}$ in the Hilbert space $\mathcal{H} = \mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{E}}$ with total hamiltonian

$$H_T = H_{\mathcal{S}} \otimes I_{\mathcal{E}} + I_{\mathcal{S}} \otimes H_{\mathcal{E}} + H_{int} \tag{21}$$

Furthermore, we assume the interaction to be of the von Neumann type (ideal).

$$H_{int} = \sum_i |s_i\rangle \langle s_i| \otimes \hat{E}_i \tag{22}$$

Initially system-environment are in seperable states as

$$|\psi\rangle = \sum_i c_i |s_i\rangle |E_0\rangle \tag{23}$$

under the unitary evolution of the whole $\mathcal{S} + \mathcal{E}$

$$\mathcal{U}(t) |s_i\rangle |E_0\rangle \longrightarrow |s_i\rangle |E_i(t)\rangle \tag{24}$$

the interaction term in the Hamiltonian dominates over the other two terms ( so that $H \simeq H_{int}$). This leads to

$$e^{-iH_{int}t/\hbar}(\sum_i c_i |s_i\rangle |E_0\rangle) \tag{25}$$

$$|\psi(t)\rangle = \sum_i c_i |s_i\rangle |E_i\rangle \tag{26}$$

Since the interaction is of von Neumann type we can say that 'the environment measures(monitors) the system'. As mentioned earlier, the superposition, initially carried by state of the system, is

now carried by the system-environment state. Density matrix of the pure state

$$\rho(t) = \sum_{i,j} c_i c_j^* (|s_i\rangle \langle s_j|) \otimes (|E_i\rangle \langle E_j|) \tag{27}$$

Our observations are restricted to the system $\mathcal{S}$ which is of our interest[5] and unfortunately, due to the large number of degrees of freedom carried by the environment (inaccessible for all practical purposes), we have to trace out the environment.

The reduced density operator to the density operator of Eq.(18) is

$$Tr_{\mathcal{E}}(\rho_{\mathcal{S}\mathcal{E}}) = \sum_{i,j} c_i c_j^* (|s_i\rangle \langle s_j|) \otimes \sum_k \langle k|E_i\rangle \langle E_j|k\rangle \tag{28}$$

Typically for macroscopically distinct states, the system-environment interaction will lead to distinguishable environmental states[1], i.e.

$$\langle E_i|E_j\rangle \longrightarrow \delta_{ij}. \tag{29}$$

therefore the system can be described by the density matrix

$$\rho_{\mathcal{S}}(t) \longrightarrow \sum_i |c_i|^2 |s_i\rangle \langle s_i| \tag{30}$$

$$\rho = \underbrace{\sum_i |c_i|^2 |\psi_i\rangle \langle \psi_i|}_{diagonal} + \underbrace{\sum_{i \neq j} c_i c_j^* |\psi_i\rangle \langle \psi_j|}_{interference} \tag{31}$$

From (21) and (22) it's easy to see that the reduced density matrix has turned diagonal and the interference terms are no longer there because the superposition of states has been destroyed locally. This reduced density matrix represents an ensemble of classical mixture. Thus, decoherence could explain why it is so difficult to observe interference on a macroscopic scale.

## 3.3 Simple model

### 3.3.1 Decoherence in two coupled harmonic oscillators

Let's model the system $'1'$ and environment $'2'$ both as single harmonic oscillators to study decoherence effect which later maybe

useful to study more complex models like the Caldeira-Leggett model.

This model has the Hamiltonian of the form

$$H_T = H_{\mathcal{S}} + H_{\mathcal{E}} + H_{int}$$

$$H_T = (\frac{\hat{p_1}^2}{2m} + \frac{k_1 \hat{x_1}^2}{2}) + (\frac{\hat{p_2}^2}{2m} + \frac{k_2 \hat{x_2}^2}{2}) + \lambda(\hat{x_1} \otimes \hat{x_2}) \tag{32}$$

(assuming interaction of only bi-linear terms)

By the following canonical transformation

$$\hat{x_1} = \sqrt{k_2}(\hat{x_1}' + \hat{x_2}') \qquad \hat{x_2} = \sqrt{k_1}(\hat{x_1}' - \hat{x_2}') \tag{33}$$

$$\hat{p_1} = 2\sqrt{k_2}\hat{p_1}' \qquad \hat{p_1} = -2\sqrt{k_1}\hat{p_2}' \tag{34}$$

the Hamiltonian becomes decoupled as

$$H_T' = (\frac{\hat{p_1'}^2}{2m} + \frac{k_1' \hat{x_1'}^2}{2}) + (\frac{\hat{p_2'}^2}{2m} + \frac{k_2' \hat{x_2'}^2}{2}) \tag{35}$$

now it is two independent QHO in terms of primed operators and we formally have the solution to this problem. By the same formalism of decoherence we will start with seperable states

$$|\psi_i\rangle = |n_1\rangle \otimes |n_2\rangle$$

After the interaction the system evolves unitarily as

$$|\psi_i\rangle \longrightarrow e^{-iH_T' t/\hbar} |n_1, n_2\rangle \tag{36}$$

$$= \sum_{n_1', n_2'} e^{-iH_T' t/\hbar} |n_1', n_2'\rangle' \langle n_1', n_2' | n_1, n_2\rangle \tag{37}$$

$$= \sum_{n_1', n_2'} e^{-iE_T' t/\hbar} |n_1', n_2'\rangle' \langle n_1', n_2' | n_1, n_2\rangle \tag{38}$$

Then by calculating reduced density matrix from here, we can study the decoherence effects.

# 4 Conclusion

Decoherence is a fundamental process that reveals the limits of quantum mechanics. Understanding open quantum systems is currently of great interest due to its implications for emerging fields such as quantum computing, quantum optics etc. Having made the efforts to study the foundation of decoherence through literature reviews led me to realize its importance in answering the basics of quantum mechanics. By having these foundations, we would like to apply new ideas to a model(Quantum Brownian motion) for the study of the decoherence effects.

## 4.1 Future directions

### 4.1.1 Study on Quantum Brownian motion

Quantum Brownian motion model consists of a particle moving in one spatial dimension and interacting linearly with an environment of independent harmonic oscillators in thermal equilibrium at a temperature $T$. Investigation on the specific scenario of extreme temperatures in the environment and the replication of the Caldeira-Leggett model's master equation for $\rho$ (reduced density matrix) using path integral techniques of quantum mechanics.

The Caldeira-Leggett model master equation[6]:

$$\frac{\partial}{\partial t}\rho(q,q',t) = \left[\frac{i}{\hbar}\left\{\frac{\partial^2}{\partial q^2} - \frac{\partial^2}{\partial q'^2} - \Omega^2(q^2 - q'^2)\right\} - \gamma(q-q')(\frac{\partial}{\partial q} - \frac{\partial}{\partial q'}) - \frac{2m\gamma k_B T}{\hbar^2}(q-q')^2\right]\rho(q,q',t). \tag{39}$$

where

$$\gamma = \frac{\epsilon^2}{4m}, \qquad \Omega = \sqrt{\Omega_0^2 - \gamma^2} \tag{40}$$

This study on the decoherence so far will be of much use for the further investigation on the model for the simultaneous decoherence of the canonical conjugates and also explore the attainment of system's classicality after the decoherence.

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